

PROVISIONAL PATENT SPECIFICATION

Method and System for Invariant-Based Fault Diagnosis Using Directive-Based Product Triples with Adaptive Nonlinearity Detection

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March 2026

1 TECHNICAL FIELD

This invention relates to methods and systems for automated fault diagnosis in electrical, industrial, and engineered systems. More particularly, the invention relates to a method of diagnosing system faults — including novel faults not present in any prior fault catalogue — by modelling the system as a three-component relational triple, computing mathematical transpositions of the triple, evaluating the closure of physical invariants across transposition frames, and identifying fault categories from the pattern and magnitude of invariant discrepancies, using an adaptive two-channel coupling kernel that automatically adjusts between linear and nonlinear analysis regimes.

2 BACKGROUND

2.1 Existing Fault Diagnosis Methods

Current methods for automated fault diagnosis in electrical and industrial systems fall into two categories.

Rule-based systems operate from a predetermined catalogue of known fault signatures. For each anticipated failure mode, a human expert encodes a diagnostic rule (e.g., “if voltage exceeds threshold X and current falls below threshold Y, then the fault is category Z”). These systems are effective for known, catalogued faults but cannot diagnose

novel or unanticipated failure modes. Every new fault type requires manual rule creation by a domain expert.

Machine-learning-based systems are trained on historical fault data to recognise patterns associated with known failure modes. These systems can generalise within the distribution of their training data but are unreliable when confronted with fault types absent from the training set. They require large labelled datasets for each fault category and must be retrained when new fault types are encountered.

Both approaches share a fundamental limitation: they diagnose by comparing observed system behaviour against a library of *previously known* fault signatures. Neither provides a systematic, first-principles method for identifying faults that have never been previously observed or catalogued.

2.2 The Problem

There exists a need for a fault diagnosis method that:

- (a) Can identify faults that have never been previously observed or catalogued;
- (b) Does not require a pre-existing fault signature library or training dataset;
- (c) Can automatically determine whether the system under diagnosis is operating in a linear or nonlinear regime;
- (d) Provides not only fault detection but fault *categorisation* from first principles;
- (e) Is applicable across multiple engineering domains (electrical, mechanical, chemical, industrial) without domain-specific redesign.

3 SUMMARY OF THE INVENTION

The present invention provides a method for diagnosing faults in an engineered system by:

1. Modelling the system as a **Directive-Based Product (DBP) triple** comprising three components: a Directive (the driving force or governing rule of the system), a Substrate (the physical medium through which the force acts), and a Product (the observable output phenomenon that exists only while the coupling between Directive and Substrate is active).

2. Defining a **coupling operation** that mathematically relates the three components through the equation $\mathcal{D} \oplus \mathcal{S} = \mathcal{P}$, where $\oplus$ is a domain-specific coupling satisfying defined formal properties including bilinearity, well-posedness, completeness, invariant preservation, and coupling dependence.
3. Computing three **transpositions** of the coupling equation — solving for the Product (forward problem), solving for the Substrate (inverse-design problem), and solving for the Directive (system identification problem) — using sensor data acquired from the system under diagnosis.
4. Evaluating the **closure of physical invariants** across all three transposition frames. In a healthy system, certain quantities (e.g., power, energy, information content) must yield identical values regardless of which transposition is used to compute them. When these invariants fail to agree, the system contains a fault.
5. Identifying the **fault category** from the specific pattern of invariant discrepancy — which invariant failed, by how much, and at which transposition — without reference to any prior fault catalogue.
6. Employing an adaptive **two-channel coupling kernel** $K = (K_{\text{spec}}, K_{\text{hol}}, \kappa)$ that automatically determines the appropriate analysis regime: a spectral channel (K_{spec}) for linear system behaviour, a holonomy channel (K_{hol}) for nonlinear system behaviour, and a nonlinearity parameter (κ) that is computed from the system data and determines the weighting between channels without manual configuration.

The invention further provides a system implementing said method, comprising sensor inputs, a computational processor, and diagnostic output means.

4 DETAILED DESCRIPTION OF THE INVENTION

4.1 Overview of the Method

The method of the invention comprises the following steps, which are described in detail in the subsections below:

Step 1: Acquire sensor data from the system under diagnosis.

Step 2: Identify and assign the three components of the DBP triple from the sensor data.

Step 3: Define the coupling operation applicable to the system domain.

Step 4: Compute the forward transposition (Solve for Product).

Step 5: Compute the first inverse transposition (Solve for Substrate).

Step 6: Compute the second inverse transposition (Solve for Directive).

Step 7: Evaluate invariant closure across all three transposition frames.

Step 8: If invariants fail to close, compute the coupling kernel to determine the analysis regime.

Step 9: Classify the fault from the pattern and magnitude of invariant discrepancies.

Step 10: Output the diagnostic result.

4.2 Step 1: Sensor Data Acquisition

The method begins by acquiring sensor measurements from the system under diagnosis. In an electrical system, this comprises at minimum voltage measurements, current measurements, and timing/phase measurements. In other engineered systems, the sensor data comprises measurements corresponding to the three components of the system triple (described in Step 2). The sensor data may be acquired in real-time for continuous monitoring or as a snapshot for point-in-time diagnosis.

4.3 Step 2: Triple Identification

The system under diagnosis is modelled as a DBP triple by identifying three components from the sensor data:

The Directive ($\mathcal{D}$): The driving force, governing rule, or input signal of the system. In an electrical system, this is typically the applied voltage. In a mechanical system, this may be the applied force or torque. In a chemical process, this may be the reactant concentration or temperature gradient. The Directive is characterised as “pure directionality” — it specifies what the system is being asked to do, but does not itself produce the observable output.

The Substrate ($\mathcal{S}$): The physical medium through which the Directive acts. In an electrical system, this is the circuit impedance (resistance, inductance, capacitance). In a mechanical system, this is the physical structure (mass, stiffness, damping). In a chemical process, this is the reaction vessel and catalyst. The Substrate exists independently of whether the coupling is active.

The Product ($\mathcal{P}$): The observable output phenomenon that exists only while the coupling between Directive and Substrate is active. In an electrical system, this is the current flow. In a mechanical system, this is the displacement or velocity. In a chemical

process, this is the reaction product. The Product ceases to exist when the coupling is deactivated (e.g., current ceases when voltage is removed from a circuit).

4.4 Step 3: Coupling Definition

The coupling operation $\oplus$ is defined for the specific system domain. The coupling must satisfy five formal properties:

Property 1 — Bilinearity: For fixed Directive, the map from Substrate to Product is linear, and vice versa.

Property 2 — Well-posedness: There exists a domain-specific condition C under which the coupling is injective in each argument (i.e., different inputs produce different outputs). In electrical systems, the well-posedness condition is that impedance is neither zero (short circuit) nor infinite (open circuit).

Property 3 — Completeness: Under the well-posedness condition, knowledge of any two components of the triple uniquely determines the third (up to equivalence class).

Property 4 — Invariant preservation: There exist quantities computable from the triple that depend only on the coupling relationship as a whole, not on which component is being solved for.

Property 5 — Coupling dependence: The Product cannot be constructed from the Directive or Substrate alone; it requires the active coupling of both.

In an electrical AC circuit, the coupling is defined by Ohm's law: $V = IZ$, where V is voltage (Directive), Z is impedance (Substrate), and I is current (Product). In nonlinear systems (e.g., semiconductor junctions), the coupling takes the form of the device characteristic equation (e.g., $I = I_0(e^{V/V_T} - 1)$ for a diode).

4.5 Step 4: Forward Transposition (Solve for Product)

Given the measured Directive and Substrate, compute the expected Product:

$$\mathcal{P}_{\text{computed}} = \mathcal{D}_{\text{measured}} \oplus \mathcal{S}_{\text{measured}}.$$

In an electrical system: given measured voltage and measured impedance, compute expected current using the coupling equation. Compare with measured current.

4.6 Step 5: First Inverse Transposition (Solve for Substrate)

Given the measured Product and measured Directive, compute the expected Substrate:

$$\mathcal{S}_{\text{computed}} = \mathcal{P}_{\text{measured}} \ominus_{\mathcal{D}} \mathcal{D}_{\text{measured}}.$$

In an electrical system: given measured current and measured voltage, compute expected impedance. Compare with measured impedance.

The inverse operation $\ominus_{\mathcal{D}}$ (stripping the Directive) is the partial inverse of $\oplus$ along the Directive axis. Its existence is guaranteed by the completeness property (Property 3) under the well-posedness condition.

4.7 Step 6: Second Inverse Transposition (Solve for Directive)

Given the measured Substrate and measured Product, compute the expected Directive:

$$\mathcal{D}_{\text{computed}} = \mathcal{P}_{\text{measured}} \ominus_{\mathcal{S}} \mathcal{S}_{\text{measured}}.$$

In an electrical system: given measured impedance and measured current, compute expected voltage. Compare with measured voltage.

The inverse operation $\ominus_{\mathcal{S}}$ (stripping the Substrate) is the partial inverse of $\oplus$ along the Substrate axis.

4.8 Step 7: Invariant Closure Evaluation

For each physical invariant φ in the invariant set I , compute φ from each of the three transposition frames:

$$\varphi_1 = \varphi(\mathcal{D}_{\text{measured}}, \mathcal{S}_{\text{measured}}, \mathcal{P}_{\text{computed}}) \tag{1}$$

$$\varphi_2 = \varphi(\mathcal{D}_{\text{measured}}, \mathcal{S}_{\text{computed}}, \mathcal{P}_{\text{measured}}) \tag{2}$$

$$\varphi_3 = \varphi(\mathcal{D}_{\text{computed}}, \mathcal{S}_{\text{measured}}, \mathcal{P}_{\text{measured}}) \tag{3}$$

In a healthy system satisfying the well-posedness condition: $\varphi_1 = \varphi_2 = \varphi_3$ for all $\varphi \in I$.

In electrical systems, the invariant set I includes: real power ($P = VI \cos \theta$), energy ($\int P dt$), phase relationship ($\theta = \arg(Z)$), and the well-posedness condition itself ($0 < |Z| < \infty$).

Fault detection occurs when any invariant fails to close: $|\varphi_i - \varphi_j| > \epsilon$ for some threshold ϵ and some $i \neq j$.

4.9 Step 8: Coupling Kernel Computation and Regime Determination

When invariant closure fails, the method computes the coupling kernel K to determine the analysis regime and refine the diagnosis.

The coupling kernel comprises three components:

The spectral channel (K_{spec}): The eigenvalues, singular values, or transfer function of the coupling viewed as a linear map. In electrical systems, this is the impedance spectrum $Z(\omega)$. This channel is computed by linearising the coupling at the measured operating point and extracting spectral data using standard linear algebra methods (eigen-decomposition, singular value decomposition, or Fourier analysis).

The holonomy channel (K_{hol}): The residue accumulated by cycling through all three transpositions in sequence (Directive $\rightarrow$ Substrate $\rightarrow$ Product $\rightarrow$ Directive). Specifically: take the measured Directive, compute the Substrate via Step 5, use that computed Substrate to compute the Product via Step 4, then use that computed Product to recompute the Directive via Step 6. Compare the recomputed Directive with the original measured Directive. The magnitude of the discrepancy is K_{hol} .

In a linear, healthy system, the cycle returns exactly to the starting value ($K_{\text{hol}} = 0$). In a nonlinear system or a system with a fault, the cycle accumulates a measurable residue ($K_{\text{hol}} > 0$). The magnitude and direction of this residue encode information about the nature and severity of the nonlinearity or fault.

The nonlinearity parameter (κ): Computed as the deviation of the measured coupling from its linearised approximation: $\kappa = \|\oplus - \oplus_{\text{lin}}\|$, where $\oplus_{\text{lin}}$ is the best linear approximation to the coupling at the measured operating point.

When κ is below a predetermined threshold (indicating the system is operating in a substantially linear regime), the diagnostic analysis relies primarily on K_{spec} . When κ exceeds the threshold (indicating significant nonlinearity), the analysis engages K_{hol} . This regime determination is automatic and requires no manual configuration.

4.10 Step 9: Fault Classification

The fault is classified from the pattern of invariant discrepancies and the coupling kernel analysis, according to the following diagnostic logic:

(a) Which invariant failed? Different invariants correspond to different physical conservation laws. A power invariant failure indicates energy is being dissipated, stored, or generated in an unaccounted manner. A phase invariant failure indicates reactive component degradation. A well-posedness condition failure indicates a fundamental structural change (approaching short or open circuit).

(b) At which transposition did the failure occur? The three transpositions probe different aspects of the system. Failure at the forward transposition (Solve for Product) indicates the output is not consistent with the input and medium — suggesting a load fault or measurement error. Failure at the first inverse transposition (Solve for Substrate) indicates the medium is not consistent with the input and output — suggesting an impedance change, component degradation, or insulation fault. Failure at the second inverse transposition (Solve for Directive) indicates the driving force is not consistent with the medium and output — suggesting a supply fault or source anomaly.

(c) What is the magnitude of the discrepancy? The size of the invariant gap quantifies the severity of the fault. Small gaps may indicate degradation or drift. Large gaps indicate acute failure.

(d) What does the coupling kernel reveal? When κ is small, the fault is occurring in a linear regime and K_{spec} identifies which spectral component (frequency, mode) is affected. When κ is large, the system is operating nonlinearly and K_{hol} characterises the coupling “curvature” — the degree to which the system’s behaviour deviates from its linearised model. The magnitude and structure of K_{hol} distinguish between different categories of nonlinear fault (e.g., saturation, hysteresis, intermittent contact, arc discharge).

The combination of (a), (b), (c), and (d) produces a **fault signature** that is specific to the fault category without requiring prior knowledge of that fault type.

4.11 Step 10: Diagnostic Output

The method outputs:

- (a) A fault/no-fault determination (binary).
- (b) If fault: the fault category as determined by the pattern analysis of Step 9.
- (c) The severity level as determined by the invariant gap magnitude.
- (d) The analysis regime (linear or nonlinear) as determined by κ .
- (e) The specific invariant(s) that failed to close, providing a starting point for physical investigation.
- (f) Optionally: the holonomy residue K_{hol} as a quantitative characterisation of the coupling nonlinearity, which may be logged for trend analysis and predictive maintenance.

5 WORKED EXAMPLES

5.1 Example 1: Power Factor Fault in an AC Motor Circuit (Linear Regime)

System: 240V single-phase AC supply feeding an induction motor rated at 1800W.

Sensor data: $V = 240$ V RMS, $I = 10$ A RMS.

Step 2 — Triple identification: $\mathcal{D} = V = 240$ V, $\mathcal{S} = Z$ (unknown), $\mathcal{P} = I = 10$ A.

Step 3 — Coupling: $V = IZ$ (Ohm's law for AC circuits).

Step 5 — Solve for Substrate: $Z = V/I = 240/10 = 24 \Omega$ (magnitude only).

Step 7 — Invariant check: Apparent power $S = VI = 2400$ VA. Expected real power from rated load: 1800 W. The power invariant does not close: $2400 \neq 1800$. Gap = 600 VA.

Step 8 — Coupling kernel: System is linear ($\kappa \approx 0$). K_{spec} reveals complex impedance with phase angle $\theta = \arccos(1800/2400) = 41.4^\circ$. Full impedance: $Z = 18 + j15.87 \Omega$.

Step 9 — Fault classification: Power invariant failure at the Solve-for-Substrate transposition. Phase relationship is nonzero. Diagnosis: **reactive power component present** (power factor = 0.75). Reactive power $Q = 1590$ VAR. Fault category: excessive reactive loading.

Step 10 — Output: Fault detected. Category: power factor degradation. Severity: moderate (power factor 0.75). Regime: linear. Recommended action: power factor correction (capacitor sizing calculable from the reactive power value).

5.2 Example 2: Nonlinear Junction Fault (Nonlinear Regime)

System: DC circuit containing a semiconductor junction.

Sensor data: $V = 0.65$ V, $I = 72$ mA.

Step 2 — Triple identification: $\mathcal{D} = V = 0.65$ V, $\mathcal{S} =$ device characteristic, $\mathcal{P} = I = 72$ mA.

Step 8 — Coupling kernel (cyclic test): Execute the full role cycle $\mathcal{D} \rightarrow \mathcal{S} \rightarrow \mathcal{P} \rightarrow \mathcal{D}$.

Station 1: Given V and device, compute $I = 72$ mA. Consistent with measurement.

Station 2: Given I and V , recover device parameters. Single operating point insufficient for unique recovery — equivalence class returned.

Station 3: With recovered parameters (finite precision), recompute V . Result: $V \approx 0.47$ V. Holonomy residue: $K_{\text{hol}} = |0.65 - 0.47| = 0.18$ V.

κ is computed as large (exponential nonlinearity). Power invariant drift: from 46.8 mW to 33.9 mW (27.6% degradation per cycle).

Step 9 — Fault classification: Large holonomy residue indicates strongly nonlinear coupling. If this system were expected to be operating linearly (e.g., in a forward-biased regime well above the knee), the holonomy magnitude diagnoses the degree of nonlinear deviation. If the holonomy exceeds expected bounds for the component, this indicates **junction degradation** or **operating point drift**.

6 SYSTEM EMBODIMENT

In one embodiment, the invention is implemented as a diagnostic system comprising:

(a) Sensor input module: One or more sensors measuring the physical quantities corresponding to the Directive, Substrate, and Product of the system under diagnosis (e.g., voltage sensors, current sensors, impedance analysers, accelerometers, pressure transducers, temperature sensors, flow meters).

(b) Computational processor: A digital processor (microcontroller, DSP, FPGA, or general-purpose computer) programmed to execute Steps 2 through 9 of the method described above. The processor receives digitised sensor data and computes the transpositions, invariant closure evaluation, coupling kernel, and fault classification.

(c) Diagnostic output module: A display, data logging system, alarm system, or communication interface that presents the diagnostic results (fault/no-fault, fault category, severity, regime, failed invariants) to a human operator or to an automated control system.

(d) Software module: Software executing the diagnostic algorithm, comprising:

- A triple definition module that maps sensor inputs to the DBP triple components based on the system domain;
- A coupling computation module that implements $\oplus$ and its inverse operations $\ominus_{\mathcal{D}}$ and $\ominus_{\mathcal{S}}$ for the relevant domain;
- A transposition engine that computes all three transpositions from sensor data;
- An invariant evaluation module that computes each invariant from each transposition frame and evaluates closure;
- A coupling kernel computation module that computes K_{spec} , K_{hol} , and κ from the system data;
- A fault classification module that analyses the invariant discrepancy pattern and coupling kernel to produce the diagnostic output.

In a further embodiment, the system operates in continuous monitoring mode, repeatedly executing the diagnostic method at a defined sampling interval and logging the coupling kernel values (particularly K_{hol} and κ) over time. Trend analysis of these values enables **predictive maintenance**: gradual increases in holonomy residue or nonlinearity parameter indicate progressive system degradation before an acute fault occurs.

In a further embodiment, the system is configured for multiple engineering domains (electrical, mechanical, chemical, industrial) by providing domain-specific implementations of the coupling operation $\oplus$ and its inverses, while the transposition engine, invariant evaluation, coupling kernel computation, and fault classification modules remain domain-independent.

7 SCOPE OF APPLICATION

The method and system of the invention are applicable to, but not limited to:

- Electrical power systems (fault diagnosis in distribution networks, motor circuits, transformer monitoring, power quality analysis, arc fault detection);
- Industrial process control (chemical process monitoring, flow system diagnosis, heat exchange efficiency monitoring);
- Mechanical systems (vibration analysis, bearing fault detection, structural health monitoring);
- Renewable energy systems (photovoltaic array fault diagnosis, wind turbine drive-train monitoring);
- Electronic systems (semiconductor device characterisation, circuit board fault isolation);
- Any engineered system that can be modelled as a three-component relational triple satisfying the five properties of the coupling operation.

8 ADVANTAGES OVER PRIOR ART

1. **Novel fault detection:** The method diagnoses faults from first principles (invariant closure) rather than comparison to a fault catalogue. It can identify faults that have never been previously observed.

2. **No training data required:** Unlike machine-learning-based diagnostic systems, the method requires no historical fault data, no labelled training sets, and no re-training when new fault types are encountered.
3. **Automatic regime determination:** The nonlinearity parameter κ automatically determines whether the system is operating in a linear or nonlinear regime and selects the appropriate analysis channel, requiring no manual configuration.
4. **Fault categorisation:** The pattern of invariant failure (which invariant, which transposition, what magnitude) provides fault categorisation, not merely fault detection.
5. **Domain independence:** The transposition engine, invariant evaluation, coupling kernel computation, and fault classification modules are domain-independent. Only the coupling operation requires domain-specific implementation.
6. **Predictive maintenance capability:** Continuous logging of the coupling kernel enables trend analysis for predictive maintenance, detecting progressive degradation before acute failure.
7. **Quantitative severity assessment:** The invariant gap magnitude provides a quantitative measure of fault severity, not merely a binary fault/no-fault determination.
8. **Self-diagnosing regime:** The framework automatically diagnoses its own analysis regime — if invariants fail to close and κ is computed, the system determines whether the failure is due to a system fault or to operating in a nonlinear regime where the holonomy channel is required.

9 INFORMAL CLAIMS

Note: Formal claims will be drafted for the complete application. The following informal claims indicate the intended scope of protection.

1. A method for diagnosing faults in an engineered system, comprising:
 - (a) acquiring sensor data from the system;
 - (b) identifying a system triple comprising a Directive, a Substrate, and a Product from the sensor data;
 - (c) computing a forward transposition and at least one inverse transposition of a coupling equation relating the three components;

- (d) evaluating closure of at least one physical invariant across the transposition frames;
 - (e) determining the presence and category of a fault from the pattern and magnitude of any invariant discrepancy.
2. The method of claim 1, further comprising computing a coupling kernel having a spectral channel and a holonomy channel, and automatically determining an analysis regime based on a nonlinearity parameter computed from the system data.
 3. The method of claim 2, wherein the holonomy channel is computed by cycling through all three transpositions in sequence and measuring the residue between the starting and ending values.
 4. The method of claim 1, wherein the fault category is determined from a combination of: which invariant failed to close, at which transposition the failure occurred, and the magnitude of the invariant discrepancy.
 5. The method of claim 1, applied to an electrical system wherein the Directive is voltage, the Substrate is impedance, and the Product is current, and the coupling equation is Ohm's law or a nonlinear device characteristic.
 6. A system for diagnosing faults in an engineered system, comprising: a sensor input module; a computational processor configured to execute the method of claim 1; and a diagnostic output module.
 7. The system of claim 6, configured for continuous monitoring and further comprising a trend analysis module that logs coupling kernel values over time to detect progressive system degradation.
 8. A computer-readable medium containing instructions that, when executed by a processor, cause the processor to perform the method of claim 1.

END OF PROVISIONAL SPECIFICATION